

ACCEPTED MANUSCRIPT • OPEN ACCESS

The influence of water condition on discharge and NO formation in an argon plasma jet

To cite this article before publication: Shuai Zhao *et al* 2026 *J. Phys. D: Appl. Phys.* in press <https://doi.org/10.1088/1361-6463/aea867>

Manuscript version: Accepted Manuscript

Accepted Manuscript is “the version of the article accepted for publication including all changes made as a result of the peer review process, and which may also include the addition to the article by IOP Publishing of a header, an article ID, a cover sheet and/or an ‘Accepted Manuscript’ watermark, but excluding any other editing, typesetting or other changes made by IOP Publishing and/or its licensors”

This Accepted Manuscript is © 2026 The Author(s). Published by IOP Publishing Ltd.



As the Version of Record of this article is going to be / has been published on a gold open access basis under a CC BY 4.0 licence, this Accepted Manuscript is available for reuse under a CC BY 4.0 licence immediately.

Everyone is permitted to use all or part of the original content in this article, provided that they adhere to all the terms of the licence <https://creativecommons.org/licenses/by/4.0>

Although reasonable endeavours have been taken to obtain all necessary permissions from third parties to include their copyrighted content within this article, their full citation and copyright line may not be present in this Accepted Manuscript version. Before using any content from this article, please refer to the Version of Record on IOPscience once published for full citation and copyright details, as permissions may be required. All third party content is fully copyright protected and is not published on a gold open access basis under a CC BY licence, unless that is specifically stated in the figure caption in the Version of Record.

View the [article online](#) for updates and enhancements.

The influence of water condition on discharge and NO formation in an argon plasma jet

Shuai Zhao* and Ana Sobota

Department of Applied Physics and Science Education, Eindhoven University of Technology, 5600 MB Eindhoven, The Netherlands

E-mail: s.zhao@tue.nl; a.sobota@tue.nl

Received xxxx

Abstract. This study investigates the effects of water condition on discharge behavior and nitric oxide (NO) production in a kHz cold argon plasma jet. Four different water configurations—floating demineralized water at room temperature, heated demineralized water, saline water, and grounded demineralized water—are compared. Laser-induced fluorescence (LIF) and optical emission spectroscopy (OES) are employed to measure ground-state NO density and excited-state NO emission intensity, respectively. Heated water promotes a more uniform discharge and the most energy-efficient local NO production, whereas saline and grounded water enhance local electron density and discharge intensity. These findings provide insights for optimizing reactive oxygen/nitrogen species generation in atmospheric-pressure plasma-liquid interactions.

Keywords: plasma jet, plasma-water interaction, discharge mode, nitric oxide, laser-induced fluorescence

The influence of water condition on discharge and NO formation in an argon plasma jet

1. Introduction

Atmospheric pressure plasma jets (APPJs) interacting with targets have attracted considerable attention in recent years [1–5]. The interaction between plasma and liquid (typically water) plays a role in determining the chemical environment at the interface, which is crucial in applications where liquid-containing targets are used [6–10].

When the plasma plume impinges on a water surface, a complex series of processes occurs, including energy transfer, charge exchange, and reactive species generation, leading to rich diversity and variations in reactive oxygen and nitrogen species (RONS) in both the gas and liquid phases. On the one hand, the water itself also influences the discharge behavior, which is responsible for many of the observed effects in plasma–liquid systems. With more entrainment of water vapor into the discharge channel through evaporation and other effects, the electron properties in plasma would change due to the increased vibrational excitation, attachment, and dissociation of water, which also affects the current value [11], the plasma development [12], the discharge morphology [13] and so on. On the other hand, different liquid conditions, such as conductivity, pH, and dissolved species, can significantly alter the plasma characteristics, which in turn affect the liquid phase [14, 15]. For example, the concentrations of RONS in plasma-activated water (PAW) were found to be significantly reduced when water temperatures are higher than 25°C [16]. Although the production rate of H_2O_2 in the solution increased, the production efficiency was found to decrease with the conductivity of the solution [17]. Therefore, understanding how liquid properties affect the plasma state is essential for controlling the chemical output both in the gas phase and the liquid phase.

Among the reactive species generated by APPJs, nitric oxide (NO) plays a critical role because of its biological activity and its involvement in plasma-induced chemical processes [18]. The production of NO in plasma jets is strongly dependent on the interaction between plasma and ambient air, as well as on the presence and properties of targets [19, 20]. In our previous work in [20], it was found that the discharge mode influenced by the presence of water is subject to the distance between jets and water. In addition, the water target significantly altered the NO formation pathway due to increased involvement of H_2O . Despite these insights, the combined effect of different water conditions on the generation of NO and the corresponding discharge dynamics remains not completely understood. In particular, how water temperature, salinity, and grounding influence these parameters has not been systematically studied.

In this work, we investigate the effect of water configuration on discharge characteristics and NO production in a kHz pulsed argon plasma jet impinging on the water. Ground-state NO density is measured using laser-induced fluorescence (LIF), and excited-state NO emission intensity is monitored using optical emission spectroscopy (OES). By comparing floating demineralized water at room temperature (the reference case) with heated, saline, and grounded water, we elucidate how water properties affect discharge characteristics and NO formation, guiding optimization of RONS production in plasma-liquid applications.

The influence of water condition on discharge and NO formation in an argon plasma jet

2. Experimental setup and methods

2.1. Plasma source and energy consumption

The Ar plasma jet in this work is the same as the one we used in previous works in [19–21], constructed in a coaxial geometry and driven by high-voltage pulses with a pulse duration of 1000 ns. The default operation parameters throughout this work are: the voltage amplitude is 5 kV, the pulse frequency is 5 kHz, and the flow rate of Ar (99.999% purity) is 1 slm. A high-voltage probe (Tektronix P6015A) and a Rogowski coil (MagneLab CT-F1.0-BNC) are used to measure the amplitude of the applied voltage V_a and the current on the high-voltage side, respectively. The discharge current I_{dis} is determined by subtracting the displacement current I_{off} (no gas flow, but voltage on) from the total current I_{on} (with gas flow and plasma). The cumulative energy consumed in the plasma from the beginning of the voltage rising edge ($t = 0$) to any given time t can be expressed as:

$$E(t) = \int_0^t V_a(t) \cdot I_{\text{dis}}(t) dt. \quad (1)$$

$E(T)$ is the total energy dissipated in one discharge cycle, and $T = 200 \mu\text{s}$ when the frequency of the voltage pulse is 5 kHz.

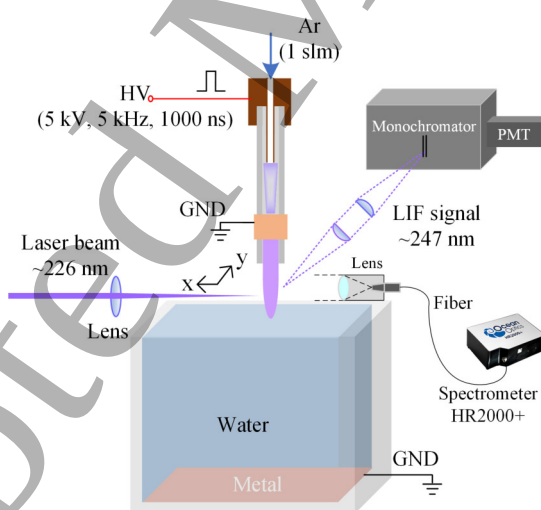


Figure 1. Schematic diagram of the experimental setup.

2.2. Water arrangement

A bulk of water is placed underneath the plasma jet, as presented in figure 1. The size of the cubic quartz container is 50 mm × 50 mm × 50 mm, and the container is filled to ensure that the surface of the water is sufficiently flat. The distance between the jet nozzle and the water surface is maintained at 10 mm. To investigate how representative water-related conditions influence discharge characteristics and NO generation in plasma-water systems, four different waters are used in this work:

The influence of water condition on discharge and NO formation in an argon plasma jet

(a) floating demineralized water at room temperature with a conductivity of around 1 $\mu\text{S}/\text{cm}$, served as a reference and denoted as ‘reference case’;

(b) floating warm water made by heating the ‘reference case’ to 55 °C, without the metallic electrode present in the vessel, denoted as ‘heated water’, to emphasize the influence of water vapor entrainment on discharge;

(c) floating homemade saline water with a concentration of 40 g/L NaCl, denoted as ‘saline water’, to stress the effect of increased electrical conductivity on discharge;

(d) demineralized water with a grounded metal plate placed at the bottom of the water, denoted as ‘grounded water’, to underline the influence of electrical boundary conditions on discharge; Note that for (a)–(c) water is floating.

Owing to the sufficiently large water volume and the short treatment duration, the electrical conductivity of the water doesn’t change significantly throughout the experiments. Moreover, the deformation of the water surface is neglected in the analysis.

The objective of this work is restricted to comparisons between the reference case and the individual heated-water, saline-water, and grounded-water cases under the specified experimental conditions.

2.3. Laser-induced fluorescence

The density of the ground-state NO, n_{gnd} , is measured *in situ* by LIF. The LIF setup and measurement method are described in detail in our previous work [19], and we only briefly summarize them here. A Nd: YAG pump laser coupled with a dye laser is used to generate the laser beam at 226.186 nm, with a 5 kHz repetition rate and a 9 ns (FWHM) pulse duration, to excite the ground-state NO($X^2\Pi$, $v = 0$) to the excited-state NO($A^2\Sigma^+$, $v = 0$). A monochromator attached with a photomultiplier is used to perpendicularly collect the fluorescence signal from the transition of NO($A^2\Sigma^+$, $v = 0$) $\rightarrow$ NO($X^2\Pi$, $v = 2$) at around 247 nm, the intensity of which is then amplified and read out by a gated photon counter. The LIF measurement is performed at the afterglow time (pulse-off time) where n_{gnd} remains constant [19,20]. The absolute value of n_{gnd} is calibrated using a gas mixture of Ar with a known concentration of NO, as described in [19].

2.4. Optical emission spectroscopy

The OES measurements are implemented with two spectrometers: the emission intensity of NO($A^2\Sigma^+$, $v = 0$) at 247 nm is measured using the same UV spectrometer that is also used for the LIF measurement; a spectrometer (Ocean Optics HR2000+) with a resolution of 0.23 nm and a slit of 10 μm is used to obtain optical spectra in the wavelength range of 415–855 nm. In the latter case, an optical fiber coupled with a collimating lens is used to collect the light emitted from the plasma plume near the water surface, as shown in figure 1.

To determine the variation of relative electron density (n_e) in the plasma plume of this Ar-dominant discharge under different water configurations, two OES-based

The influence of water condition on discharge and NO formation in an argon plasma jet

methods that have been proposed in [22] are used: (a) A line-ratio method: the ratio of two Ar lines (706.7 nm/696.5 nm) is adopted to assess the trend of n_e , given that the atom quenching coefficient of the upper level of 706.7 nm (Ar 2p₃) is much larger than that of the upper level of 696.5 nm (Ar 2p₂), while their electron quenching coefficients as well as excitation rate coefficients are similar. Thus, the intensity ratio of 706.7 nm/696.5 nm is positively correlated with n_e . (b) A continuum emission method: three intervals from pure continuum emission, namely 440–450 nm, 460–470 nm and 590–600 nm, are selected to confirm the trend of n_e , since continuum emission is mainly formed by the collisions of remaining electrons with atoms and the intensity of continuum emission is found to be proportional to the square of n_e [23]. More details about these two methods can be found in [22]. It should be noted that the Ar line ratio and continuum emission intensity are employed here as relative indicators of electron-density variation rather than direct measurements of absolute electron density.

3. Experimental results and discussion

3.1. Discharge phenomena

Figures 2(a)-(d) present the discharge morphology when the plasma jet impinges on different water targets. Figure 3 displays the I-V plots corresponding to the discharges in figure 2. As seen in figure 2(a), the plume presents a multi-filament feature when the reference case is used, which has been reported in [20]. However, heated water enables more homogeneous but less intense discharge, as shown in figure 2(b). The discharge current for the heated water case is lower than that of the reference case, while the voltage applied to the high-voltage pin electrode stays unchanged, implying that the equivalent resistance of the plasma is larger when the water is preheated. This can be explained by the existence of more H₂O molecules in the plume due to faster evaporation of water from the surface, as measured in [21]. H₂O promotes more low-threshold excitation and vibrational channels, increasing the collision rate and effective collision cross section, which reduces electron mobility and consequently decreases plasma (or electron) conductivity [24,25]. As a result, the current density is suppressed, making it difficult to sustain filamentary channels, and the discharge looks more diffuse. This can be further supported by the variation of relative n_e that will be discussed in the following sections.

Figure 2(c) indicates that saline water leads to more intense filaments. A higher conductivity of water enables the rapid penetration of charges once they come into contact with the water surface. The resistance of the water bulk that is in series with the plasma channel is calculated to be about 3 Ω, which is negligible compared to that of the reference case, 200 kΩ. The equivalent resistance of the water was calculated based on its conductivity and volume. The discharge current then becomes higher in this case, as can be seen in figure 3. The grounded water in figure 2(d) triggers a similar intense discharge to the saline water. The grounded metal plate at the bottom of the water

The influence of water condition on discharge and NO formation in an argon plasma jet

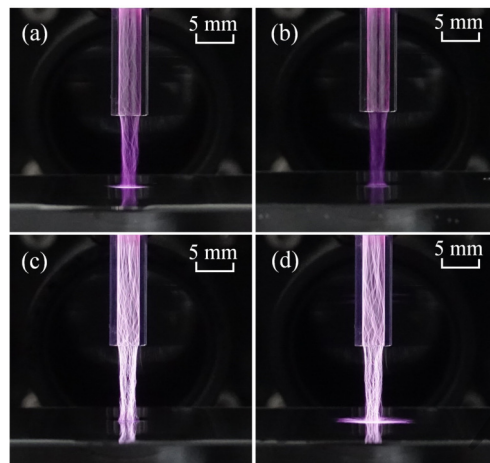


Figure 2. Discharge morphology of plasma jet impinging on different water targets. (a)-(d) respectively correspond to the cases of reference, heated water, saline water, and grounded water. The distance between the capillary nozzle and the water surface was kept at 10 mm. The discharge was performed with an applied voltage of 5 kV and an Ar gas flow of 1 slm in all cases. The images were taken with a digital camera with an exposure time of 1/60 s.

provides a directional release path for the charges, which lowers the total equivalent resistance of this plasma-water system, in contrast with the floating ones. However, unlike figure 2(c) which displays a constricted contact area on the water surface, figure 2(d) presents an expanded discharge area on the plasma-water interface. This is because the surface of grounded water has a longer charge relaxation time as a result of its low conductivity, which enables the charges to expand laterally. Still, acting as a floating electrode, the surface potential of floating saline water is easily elevated when the plasma touches the surface [26], because of its high conductivity. Horizontally, smaller surface potential differences lead to a restricted discharge area.

3.2. Dissipated energy

The discharge current difference presented in figure 3 consists of two parts, separated by t_1 ($t_1 \approx 230$ ns), when the plasma ionization wave (IW) is just about to touch the water surface. iCCD imaging presenting the IW propagation can be found in our previous work [20], and it is not shown here. Note that the oscillation of the discharge current is due to the existence of an equivalent inductor in our system, forming an underdamped RLC oscillation with the resistor and capacitor in the circuit [27].

Accordingly, the energy consumption in discharge can be divided into two categories: before the IW touches the water and the whole discharge cycle, as shown in figure 4. It can be seen that the energy dissipation before the plasma touches the water is very close for all cases. However, the total energies consumed throughout the discharge cycle are quite different for varied water conditions. This means that a significant difference in energy consumption occurs at a duration later than t_1 . In general, after

The influence of water condition on discharge and NO formation in an argon plasma jet

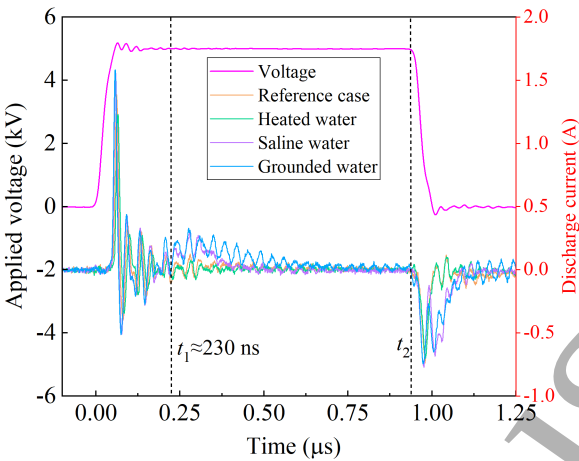


Figure 3. Voltage-discharge current waveforms (I-V plots) for plasma jet impinging on different water targets. The discharge was performed at 5 kV with 1 slm. t_1 represents the moment when the plasma ionization wave first reaches the water surface.

t_1 , the plasma immediately forms a secondary IW or so-called return stroke [20, 26], expanding the discharge duration. The energy consumption gradually increases within this pulse-on time till the falling edge discharge arrives to form a negative current to compensate for some dissipation from t_2 . The energy compensation at the falling edge is usually lower than the energy dissipated between t_1 and t_2 , which are the cases of the reference, saline water, and grounded water. However, heated water leads to a lower total energy consumption compared to the energy consumed before t_1 , indicating that, in this scenario, the energy dissipated in the duration between t_1 and t_2 is less than that compensated at the falling edge discharge. Saline water and grounded water greatly increase total energy dissipation compared to the reference case, as a result of a higher and longer-duration discharge current between t_1 and t_2 , as shown in figure 3. This can be correlated with the discrepancy in NO density observed in the following section.

3.3. Average NO density

In our previous work [20], in the presence of a water target, it was found that the variation of the NO density (n_{gnd}) was minimal close to the jet nozzle and maximal near the water surface, with a gradual transition along the axial axis. Therefore, to compare the n_{gnd} change in this work, two measurement points are selected: 1 mm below the jet nozzle and 2 mm above the water surface.

Figure 5 shows n_{gnd} under various water conditions. It can be seen that, generally, n_{gnd} exhibits similar relations for both locations: compared to the reference case, heated water lowers n_{gnd} , while both saline water and grounded water elevate n_{gnd} . n_{gnd} at 1 mm below the nozzle varies within a narrow range. The result for 2 mm above the water is a rough amplification of that for 1 mm below the jet nozzle. n_{gnd} also roughly follows the trend as the total energy consumption in figure 4, since the production of important reactive species (N, O) determining NO generation directly depends on the impact of

The influence of water condition on discharge and NO formation in an argon plasma jet

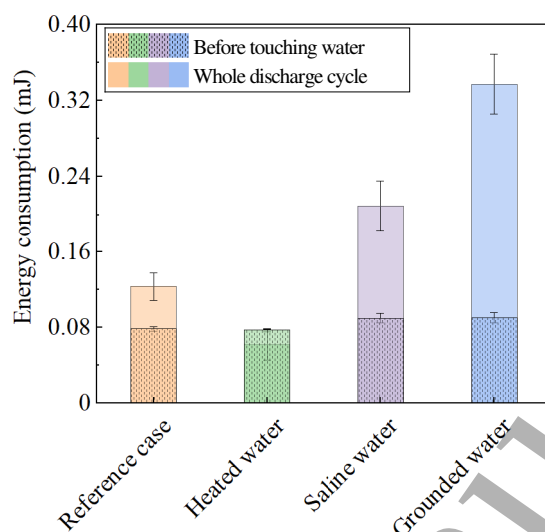


Figure 4. Energy consumption in the discharge for different water configurations: before the ionization wave touches the water and for the whole discharge cycle. The discharge was performed at 5 kV with 1 slm Ar.

electrons, whose amount is dependent on the energy injected throughout the discharge. More specifically, the energy dissipation between t_1 and t_2 in figure 3 contributes to most of the difference in NO production in the vicinity of the water surface. The improvement of n_{gnd} for the heated water case is minor compared to the others, indicating an axially uniform distribution of n_{gnd} in this case.

To calculate the amount of NO produced per J (Joule), the local apparent NO

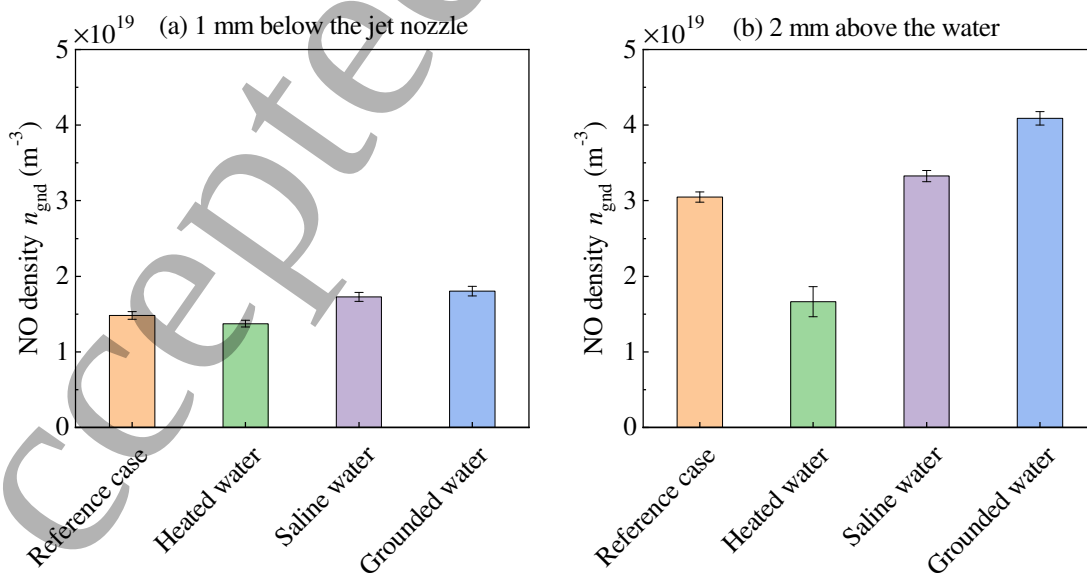


Figure 5. Ground-state NO density (n_{gnd}) respectively measured at (a) 1 mm below the jet nozzle and (b) 2 mm above the water surface. The discharge was performed at 5 kV with 1 slm Ar.

The influence of water condition on discharge and NO formation in an argon plasma jet

production efficiency η_{NO} is introduced. The calculation can be done via the expression:

$$\eta_{\text{NO}} = \frac{n_{\text{gnd}} \cdot Q}{N_{\text{a}} \cdot P} \tag{2}$$

where n_{gnd} is the local NO density (in a unit of m^{-3}) measured by LIF, Q is the flow rate of the working gas (in a unit of $\text{m}^3 \cdot \text{s}^{-1}$), N_{a} (mol^{-1}) is the Avogadro constant, and P is the discharge power (in a unit of $\text{J} \cdot \text{s}^{-1}$) that can be calculated through $P = E(T)/T$.

η_{NO} for different scenarios are shown in table 1. The results indicate that in all cases the water facilitates a higher NO production efficiency near the interface. This may be due to the enhanced involvement of H_2O molecules near the water surface, which may increase the availability of OH-related species that can contribute to NO production through the reaction pathway $\text{N} + \text{OH} \rightarrow \text{NO} + \text{H}$. The importance of this pathway has been discussed in detail in [20]. Higher energy consumption does not necessarily increase NO production per J and may even lower η_{NO} , as observed for saline water and grounded water. Instead, heated water with the lowest energy dissipation results in the highest η_{NO} . Possible explanations are as follows. First, a higher water temperature promotes water evaporation and enhances the mixing of H_2O molecules with the working gas, providing additional reactants for NO formation. Second, the presence of H_2O may enhance vibrational collisions and reduce the fraction of high-energy electrons, leading to a shorter high-energy tail and a broader low-energy region in the electron energy distribution function (EEDF). This may in turn facilitate the dissociation of H_2O into OH and H, since this process requires a relatively low threshold energy of 5.17 eV [28]. In addition, the reduction of η_{NO} at higher power may also be related to the limited availability of N_2 and O_2 , whose densities in the present work depend on the degree of air entrainment.

Table 1. Local apparent NO production efficiency η_{NO} (10^{-10}mol/J) for different water conditions

Measurement point	Reference case	Heated water	Saline water	Grounded water
1 mm below jet nozzle	6.7	12.4	4.6	3.0
2 mm above water surface	13.8	15.0	8.8	6.7

The column bars in figure 6 present the intensities of OES emission at 247 nm integrated in the entire discharge cycle, for various water targets, measured at 2 mm above the interface. It can be seen that it generally follows the trends of energy consumption in figure 4 and n_{gnd} in figure 5. The excited NO molecules can be either from the ground-state NO generated when the plasma IW arrives to provoke new reactions or the long-lived leftover NO generated in the past discharge cycles. The relative ratio of the OES intensity to the ground-state NO density is calculated, as shown by the dotted line in figure 6. It can be seen that the ratios for the cases of saline water and grounded water are comparable, while for the heated water, it is rather low.

The influence of water condition on discharge and NO formation in an argon plasma jet

The low excitation rate could be a result of the low electron density, as indicated in figure 7, and possibly also of the low electron temperature in this case.

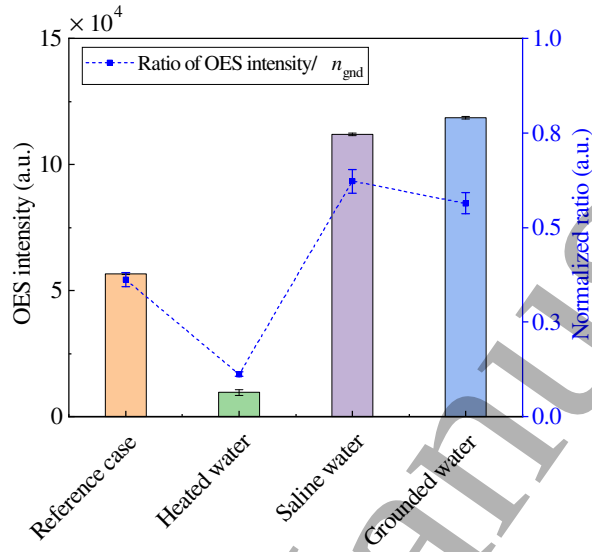


Figure 6. The OES emission intensities at 247 nm (column bars) and the relative ratio (dotted lines) of OES intensity to the n_{gnd} for different water configurations. The measurement point was 2 mm above the water surface. The OES intensity was integrated for the whole discharge cycle, and the n_{gnd} used here was from the result in figure 5, obtained in the afterglow time. The discharge was performed at 5 kV with 1 slm Ar.

The trend of relative electron density n_e is shown in figure 7. The results obtained from both the line ratio method and the continuum emission method are in good agreement. In addition, relative n_e shares a similar trend as the excitation portion in figure 6, particularly a lower value for the case of heated water. This indicates that, in this work, the excitation process $\text{NO}(\text{X}^2\Pi, v=2) \rightarrow \text{NO}(\text{A}^2\Sigma^+, v=0)$ highly depends on electron impact. In [18], van Gessel *et al* concluded that the most likely excitation mechanism is through impact with the $\text{N}_2(\text{A})$ metastable, while they conducted the measurement in an RF discharge. Van Rooij *et al* [29] found that the preheated water led to an increase in absolute electron density measured using Stark broadening, while they experimented with a pin-water setup in air, where the discharge was very different, drawing much more current and going into a different mode of operation, and the gas temperature during the discharge was much higher than that in this work.

A higher energy injection accompanied by a slightly lower relative n_e is observed in the grounded water case compared to the saline water case. When the demineralized water is grounded, the discharge expands over a larger area, as explained in section 3.1. The broader current path leads to a slightly higher total current but a lower local n_e due to the enlarged conductive cross-section. In contrast, as mentioned in section 3.1, when the saline solution is electrically floating, the accumulated surface charge rapidly modifies the interfacial potential, leading to a local field concentration and a narrower,

The influence of water condition on discharge and NO formation in an argon plasma jet

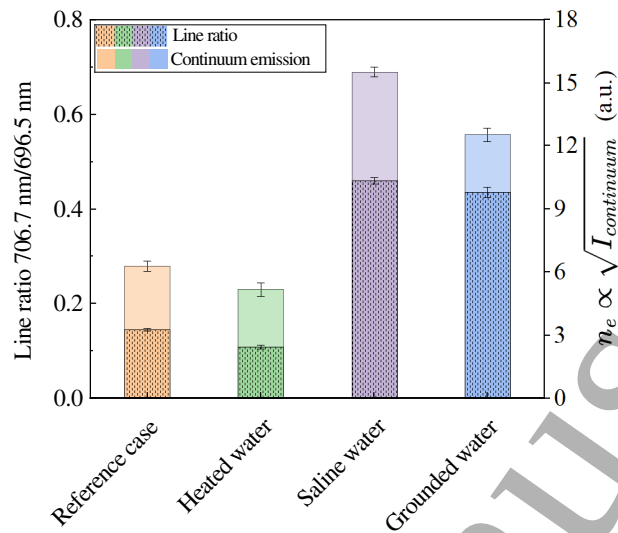


Figure 7. Relative electron densities n_e in the cases of different water configurations deduced from the line ratio method and the continuum emission method. The discharge was performed at 5 kV with 1 slm Ar.

more intense discharge channel, yielding higher local n_e but smaller total current. This could also be one of the reasons that account for the difference in η_{NO} shown in table 1. The n_e difference measured under different grounding conditions in plasma-liquid interactions was also observed in [30].

3.4. Time-resolved NO($A^2\Sigma^+$) emission

The time-resolved OES intensities from the transition NO($X^2\Pi$) $\rightarrow$ NO($A^2\Sigma^+$) at 247 nm, measured at 2 mm above the interface, are displayed in figure 8. Their peak times are not identical. Besides, compared to the reference case and heated water, both saline water and grounded water result in an earlier appearance of the OES signal and a broader duration in both the rising edge and falling edge discharges. All of these suggest that the water condition influences the propagation of the IW, both velocity and profile.

A second emission peak, following immediately after the first one, can be observed for the reference case, which is attributed to the return stroke and has been found in different target conditions, such as metal, distilled water, and biological tissue [12,31–33]. This is not clearly seen in the cases of saline water and grounded water in figure 8, likely due to the overlapping of peaks caused by the intense, long-duration discharge. Once the IW touches the water, charges are rapidly and continuously either neutralized or released by saline water or grounded water, maintaining a longer decay of emission. Although the OES peak intensity for saline water exceeds that for grounded water, its subsequent decay occurs more rapidly. Note that the energy consumption of saline water is lower than that of grounded water; in this case, a higher peak is another evidence of a higher local instantaneous n_e .

The influence of water condition on discharge and NO formation in an argon plasma jet

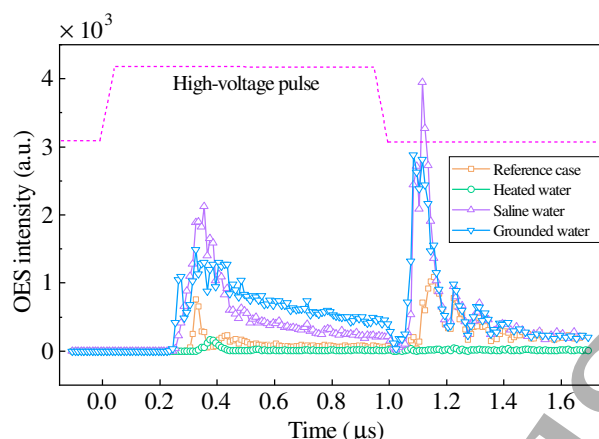


Figure 8. Time-resolved $\text{NO}(\text{A}^2\Sigma^+)$ emission (at 247 nm) intensity at 2 mm above the target surface with different water configurations. The time interval between data points was 20 ns. The discharge was performed at 5 kV with 1 slm Ar.

It should be noted that the water condition grouping in this work may not constitute a fully orthogonal parameter study. For instance, in the heated water case, factors such as gas temperature and water-vapor entrainment may vary simultaneously and therefore cannot be completely decoupled. In addition, the current setup intends to conduct comparisons between representative plasma-water configurations and does not include full ranges of water temperature and salinity. Future work will involve more systematic investigations with independent variation of these parameters to better distinguish their respective contributions.

4. Conclusions

In this work, the influence of different water configurations on the discharge behavior and nitric oxide (NO) production efficiency has been investigated in a kHz pulsed atmospheric-pressure Ar plasma jet. It is found that different water targets—floating demineralized water at room temperature (the reference case), heated demineralized water, saline water, and grounded demineralized water—significantly influence the discharge morphology, energy dissipation, relative electron density and NO production.

Heated water leads to a more homogeneous but less intense discharge due to more H_2O evaporating into the plume, whereas saline water and grounded water result in stronger filaments and longer effective discharge duration due to the fast neutralization and release of charges through the water. The energy consumption during one discharge cycle follows: heated water < reference case < saline water < grounded water, which is also in line with the trend of their corresponding ground-state NO densities, n_{gnd} . However, the local apparent efficiency in NO generation, i.e., the amount of NO produced per J, is found to be heated water > reference case > saline water > grounded water, which can be attributed to the differences in OH radicals offered by H_2O dissociation, the possible variation of the electron energy, and the limited amount of source reactant

*The influence of water condition on discharge and NO formation in an argon plasma jet*¹³
for NO formation.

Overall, these results demonstrate that water conditions not only influence the discharge dynamics but also play a critical role in controlling NO generation. The findings provide valuable insights for optimizing plasma-liquid interactions in applications where selective NO production is desired, and they contribute to a deeper understanding of the interplay between plasma discharge characteristics and reactive oxygen/nitrogen species formation in atmospheric-pressure plasma jets.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgements

This work is partially funded by the TU/e-UU-UMCU Alliance program. The authors sincerely thank Pieter Sanders and John Meulendijks for their assistance with the experiments.

References

[1] Jiang Y, Wang Y, Zhang J and Wang D 2023 *Journal of Physics D: Applied Physics* **56** 085201
[2] Sobota A, Garcia-Caurel E and Guaitella O 2024 *Journal of Physics D: Applied Physics* **58** 063005
[3] He G, Liu Y, He F, Miao J, Li J, Zhang Y, Gao Z, Wang R, Yan X and Ouyang J 2024 *Physics of Plasmas* **31** 083503
[4] Martinez L, Dhruv A, Lin L, Balaras E and Keidar M 2019 *Plasma Sources Science and Technology* **28** 115002
[5] Uchida G, Nakajima A, Takenaka K, Koga K, Shiratani M and Setsuhara Y 2015 *IEEE Transactions on Plasma Science* **43** 4081–4087
[6] Bruggeman P J, Frontiera R R, Kortshagen U, Kushner M J, Linic S, Schatz G C, Andaraarachchi H, Chaudhuri S, Chen H T, Clay C D *et al.* 2025 *The Journal of Chemical Physics* **162** 071001
[7] Bruggeman P J, Kushner M J, Locke B R, Gardeniers J G, Graham W, Graves D B, Hofman-Caris R, Maric D, Reid J P, Ceriani E *et al.* 2016 *Plasma sources science and technology* **25** 053002
[8] Stancampiano A, Simoncelli E, Boselli M, Colombo V and Gherardi M 2018 *Plasma Sources Science and Technology* **27** 125002
[9] Lai M, Song S, Oshin E, Potter L, Lai N and Jiang C 2022 *Journal of Applied Physics* **131** 173301
[10] Wang H, Wandell R J, Tachibana K, Voráč J and Locke B R 2018 *Journal of Physics D: Applied Physics* **52** 075201
[11] Luo Y, Lietz A M, Yatom S, Kushner M J and Bruggeman P J 2018 *Journal of Physics D: Applied Physics* **52** 044003
[12] Yang Y, Zhang Y, Liao Z, Pei X and Wu S 2017 *IEEE Transactions on Radiation and Plasma Medical Sciences* **2** 223–228
[13] Du Y, Nayak G, Oinuma G, Peng Z and Bruggeman P J 2017 *Journal of Physics D: Applied Physics* **50** 145201
[14] Hamdan A, Profili J and Cha M S 2020 *Plasma Chemistry and Plasma Processing* **40** 169–185
[15] Yoon S Y, Jeon H, Yi C, Park S, Ryu S and Kim S B 2018 *Scientific reports* **8** 12037

The influence of water condition on discharge and NO formation in an argon plasma jet

- [16] Pang B, Liu Z, Zhang H, Wang S, Gao Y, Xu D, Liu D and Kong M G 2022 *Plasma Processes and Polymers* **19** 2100079
- [17] Oshin E A, Rahman M Z and Jiang C 2025 *Physica Scripta* **100** 075606
- [18] Van Gessel A, Alards K and Bruggeman P 2013 *Journal of Physics D: Applied Physics* **46** 265202
- [19] Zhao S, van de Wege R and Sobota A 2025 *Plasma Sources Science and Technology* **34** 045013
- [20] Zhao S, van de Wege R and Sobota A 2026 *Plasma Sources Science and Technology* **35** 025009
- [21] Zhao S and Sobota A 2026 *Physics of Plasmas* **33** 010702
- [22] Zhao S, Wang Y, Zhu X and Sobota A 2024 *Journal of Physics D: Applied Physics* **57** 265204
- [23] Lissovski A A and Treshchalov A B 2006 Kinetics of vuv-vis spontaneous emission of high-current pulsed volume discharge in argon *Atomic and Molecular Pulsed Lasers VI* vol 6263 (SPIE) pp 110–119
- [24] Yoon J S, Song M Y, Han J M, Hwang S H, Chang W S, Lee B and Itikawa Y 2008 *Journal of Physical and Chemical Reference Data* **37** 913–931
- [25] Itikawa Y and Mason N 2005 *Journal of Physical and Chemical reference data* **34** 1–22
- [26] Schalk M, Ryan C, Coppus B, Zhao S and Sobota A 2025 *Journal of Physics D: Applied Physics* **58** 355207
- [27] Monge-Dauge C, Clement F, Svarnas P, Loiseau J F, Ricard A and Held B 2012 *IEEE Transactions on Plasma Science* **40** 2254–2260
- [28] Linstrom P J and Mallard W G Nist chemistry webbook, nist standard reference database number 69 <https://webbook.nist.gov/chemistry/>
- [29] van Rooij O, Ahlborn O and Sobota A 2024 *Journal of Physics D: Applied Physics* **57** 385206
- [30] Slikboer E and Walsh J 2021 *Scientific reports* **11** 17749
- [31] Riès D, Dilecce G, Robert E, Ambrico P, Dozias S and Pouvesle J M 2014 *Journal of Physics D: Applied Physics* **47** 275401
- [32] Darny T, Pouvesle J M, Puech V, Douat C, Dozias S and Robert E 2017 *Plasma Sources Science and Technology* **26** 045008
- [33] Babaeva N Y, Naidis G V, Panov V A, Wang R, Zhang S, Zhang C and Shao T 2019 *Plasma Sources Science and Technology* **28** 095006